



A New Version of the Generalized Zhang–Zhu Strength Criterion and a Discussion on Its Smoothness and Convexity

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Abstract

The generalized Zhang–Zhu (GZZ) strength criterion has a high associated fitting accuracy for true triaxial test data. It makes direct use of the parameters of the Hoek–Brown criterion and has become one of the strength criteria recommended by the International Society for Rock Mechanics (ISRM). In this study, a new version of the GZZ criterion is developed by addressing the problems associated with the original version's non-smoothness and non-convexity on the π plane. (i) The new criterion is smooth and derivable on the π plane with no numerical singularity in the strength calculation. It thus renders the application of complicated Lode dependence functions for modification of the non-smooth GZZ criterion unnecessary. (ii) The new criterion is non-convex only under the triaxial extension stress condition, which is not critical for those criteria developed from the Hoek–Brown strength criterion. The new criterion is transformed from incomplete convexity to full convexity with an increase in hydrostatic pressure I_1 . Additionally, there exists a critical transformation radius r_0 , which only relates to I_1 . The new criterion achieves full convexity when I_1 is high and the quality of the rock mass is poor. (iii) Both the new criterion and the GZZ criterion have high fitting accuracy for rock strength data. Their fitting accuracy is much higher than that of the Pan–Hudson criterion. However, the fitted strength parameter m of the new criterion is closer to Hoek's recommendation than that of the GZZ criterion, thus proving that the new criterion is applicable and reliable when used for rock strength calculation based on the Hoek–Brown criterion parameter system.

Keywords GZZ strength criterion · New GZZ criterion · Smooth and convex · Parameter system · Fitting precision · Applicable for rock mass

List of symbols

$\sigma_1, \sigma_2, \sigma_3$	Maximum, intermediate principal stress and minimum principal stress
σ_i^e	Effective principal stress
I_1, I_2, I_3	First, second and third stress invariant
J_2, J_3	Second and third deviatoric stress invariant
θ_σ	Lode angle
σ_c	Unconfined compressive strength (UCS)
m_i	Material constant for intact rock
m_b, s, a	Strength parameters for rock masses
τ_{oct}	Octahedral shear stress

$\sigma_{m,2}$	Effective mean stress
$(\sigma_{m,2})_{new}$	Effective mean stress of new GZZ
$(\sigma_{m,2})_{GZZ}$	Effective mean stress of original GZZ
S_{ij}	The deviatoric stress tensor
ξ	Hydrostatic stress axis
$r(\theta_\sigma)$	Radius of strength envelope to hydrostatic stress axis
r_0	Critical radius of the new criterion from non-full convex to full convex on the π plane
β	Ratio of $r(\pi/6)$ to $r(-\pi/6)$
mf	Evaluation index of absolute error (SAE) method
R^2	Square of correlation coefficient

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1 Introduction

The Hoek–Brown strength criterion is widely used for rock mass engineering stability analysis (Hoek and Brown 2019). Its parameters can be comprehensively obtained by

performing triaxial compression tests and examining mineral compositions, discontinuity distributions, and geological features (Hoek and Brown 2019; Zhang and Zhu 2007; Chen et al. 2021b). Based on many triaxial tests and engineering field test data, Hoek et al. performed a significant amount of meaningful research on criterion parameters and developed a complete and reliable method for obtaining the criterion parameters (Hoek and Brown 1988, 1997, 2019; Hoek et al. 2000, 2002; Marinos and Hoek 2001). However, the Hoek–Brown criterion is a two-dimensional criterion not considering the intermediate principal stress σ_2 , which has a significant influence on rock strength, especially for rocks under low-stress conditions (Al-Ajmi and Zimmerman 2005; Chang and Haimson 2000; Colmenares and Zoback 2002; Mogi 1967, 1971b; Pan and Hudson 1988). Therefore, some three-dimensional Hoek–Brown criteria have been developed in recent years for the analysis of the strength characteristics of rock mass under various stress conditions, resulting in three-dimensional Hoek–Brown strength criteria (Jiang 2017a, b; Jiang et al. 2011; Jiang and Xie 2012; Jiang and Zhao 2015; Pan and Hudson 1988; Priest 2005; Zhang 2008; Zhang and Zhu 2007). Among these, the generalized Zhang–Zhu (GZZ) strength criterion demonstrates high fitting precision for true triaxial test data. More importantly, it degenerates into the Hoek–Brown criterion in triaxial compression and extension stress conditions, making it possible to reliably use the Hoek’s recommendation regarding the Hoek–Brown strength parameters (Zhang 2008; Zhang and Zhu 2007).

However, the GZZ criterion, like the Hoek–Brown criterion, is non-smooth and non-convex on the π plane, which may result in numerical singularity and non-convergence in strength calculations of rock mass (Bardet 1990; Jiang and Pietruszczak 1988; Lin and Bazant 1986). To satisfy the requirements of full smoothness and convexity, various studies have explored its mathematical requirements and developed various Lode dependences (Bardet 1990; Matsuoka and Nakai 1974; Willam 1975; Yu 1992). Zhang et al. (2013) used three different Lode dependences to modify the GZZ criterion and established the modified GZZ strength criterion. However, the modified criterion involves complicated mathematics with a complicated iterative process for calculating the strength of rock mass.

In our study, we developed a new version of the GZZ criterion that possessed full smoothness on the π plane. In the following sections, we discuss its geometric convexity for rock masses of different quality and compare the fitting precision of the new criterion, the original GZZ criterion, and the Pan–Hudson criterion to analyze the fitting accuracy and parameter reliability of the newly-developed criterion.

2 Hoek–Brown Criterion and Its Existing 3D Versions

The strength criterion possesses different expressions in different stress spaces. The main expressions include $F(\sigma_1, \sigma_2, \sigma_3)=0$ in the principal stress tensor, $F(I_1, I_2, I_3)=0$ in the stress invariant, and $F(I_1, J_2, \theta_\sigma)=0$ on the π plane in terms of the hydrostatic stress I_1 and Lode angle θ_σ . In these expressions, $\sigma_i (i=1, 2, 3)$ is the stress tensor; $I_i (i=1, 2, 3)$ is the stress invariant; $I_1=\sigma_{ii}=\sigma_1+\sigma_2+\sigma_3$; $I_2=\sigma_1\sigma_2+\sigma_2\sigma_3+\sigma_1\sigma_3$; $I_3=\det(\sigma_{ij})=\sigma_1\sigma_2\sigma_3$; J_2 is the second deviatoric stress invariant, and $J_2=S_{ij}S_{ji}/2=(I_1^2-3I_2)/3$. The main form used in this paper is $F(I_1, J_2, \theta_\sigma)=0$ on the π plane, which is convenient for the study of the geometric properties of a strength criterion. The principal stress space with its transform relationship on the π plane is illustrated in Fig. 1.

Figure 2 shows the Lode angle on the π plane in the coordinate system of $\sigma_i (i=1, 2, 3)$, which reflects the influence of σ_2 on rock strength.

$$\theta_\sigma = \tan^{-1} \left[\frac{2\sigma_2 - (\sigma_1 + \sigma_3)}{\sqrt{3}(\sigma_1 - \sigma_3)} \right]. \quad (1)$$

The stress tensor σ_i can be expressed in terms of I_1, J_2 , and θ_σ as follows:

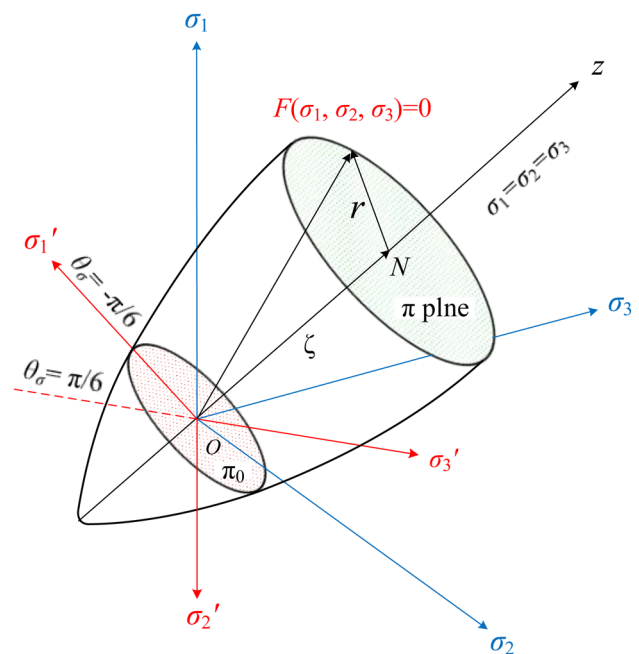


Fig. 1 Principal stress space on the π plane

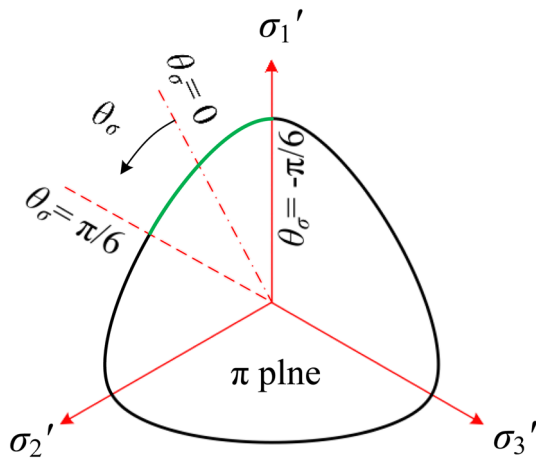


Fig. 2 Lode angle on the π plane

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \end{bmatrix} = \frac{2}{\sqrt{3}} \sqrt{J_2} \begin{bmatrix} \sin\left(\theta_\sigma + \frac{2}{3}\pi\right) \\ \sin(\theta_\sigma) \\ \sin\left(\theta_\sigma - \frac{2}{3}\pi\right) \end{bmatrix} + \begin{bmatrix} I_1/3 \\ I_1/3 \\ I_1/3 \end{bmatrix}. \quad (2)$$

2.1 Hoek–Brown Strength Criterion

Hoek and Brown proposed the original Hoek–Brown strength criterion for intact rock in 1980 (Hoek and Brown 1980a, b). It reflects the nonlinear relationship between σ_1 (rock strength) and σ_3 (confining pressure) on rock failure as follows:

$$\sigma_1 = \sigma_3 + \sigma_c \left(m_i \frac{\sigma_3}{\sigma_c} + 1 \right)^{0.5}, \quad (3)$$

where σ_c is unconfined compressive strength, and m_i is the material constant for intact rock.

Considering the limitations and deficiencies of the original criterion, Hoek et al. (2000) and (2002) proposed the general Hoek–Brown criterion. This expression is suitable for various types of rocks and rock mass and is expressed as follows:

$$\sigma_1 = \sigma_3 + \sigma_c \left(m_b \frac{\sigma_3}{\sigma_c} + s \right)^a, \quad (4)$$

where m_b , s , and a are strength parameters for rock masses, and s varies from 0 to 1 (totally broken rock mass to complete rock mass, i.e. rock). The general Hoek–Brown strength criterion degenerates to the original when $s=1$ and $a=0.5$. Currently, the Hoek–Brown criterion with its recommended parameters for rock masses is widely accepted and has been applied to many projects around the world (Hoek

and Brown 1988, 1997, 2019; Hoek et al. 2000, 2002; Mariños and Hoek 2001).

By substituting Eq. (2) into the Hoek–Brown criterion (Eq. 4), we can derive the expression for the Hoek–Brown criterion on the π plane as follows:

$$\frac{1}{(\sigma_c)^{(1/a-1)}} \left(2 \cos \theta_\sigma \sqrt{J_2} \right)^{1/a} + \left(\cos \theta_\sigma + \frac{\sin \theta_\sigma}{\sqrt{3}} \right) m \sqrt{J_2} - m \frac{I_1}{3} - s \sigma_c = 0. \quad (5)$$

Noting that $J_2 = 3\tau_{oct}^2/2$, the Hoek–Brown criterion can also be written as follows:

$$\frac{1}{(\sigma_c)^{(1/a-1)}} \left(\sqrt{6} \cos \theta_\sigma \tau_{oct} \right)^{1/a} + \left(\frac{\sqrt{6} \cos \theta_\sigma}{2} + \frac{\sin \theta_\sigma}{\sqrt{2}} \right) m \tau_{oct} - m \frac{I_1}{3} - s \sigma_c = 0 \quad (6)$$

2.2 Pan–Hudson Criterion

Based on the Hoek–Brown criterion, Pan and Hudson developed a three-dimensional strength criterion called the Pan–Hudson criterion (Pan and Hudson 1988):

$$\frac{3}{\sigma_c} J_2 + \frac{\sqrt{3}}{2} m \sqrt{J_2} - m \frac{I_1}{3} - s \sigma_c = 0. \quad (7)$$

By substituting $J_2 = 3\tau_{oct}^2/2$ into Eq. (7), we can derive:

$$\frac{9}{2\sigma_c} \tau_{oct}^2 + \frac{3}{2\sqrt{2}} m \tau_{oct} - m \frac{I_1}{3} - s \sigma_c = 0 \quad (8)$$

The Pan–Hudson criterion is circular on the π plane with a low fitting accuracy to the true triaxial test data and will be further discussed later.

2.3 Priest Criterion

Priest developed a three-dimensional Hoek–Brown criterion by combining the Hoek–Brown criterion with the Drucker–Prager criterion, which yields (Priest 2005):

$$\sqrt{J_2} - \frac{A(\sigma_1^e + \sigma_2^e + \sigma_3^e)}{3} - B = 0, \quad (9)$$

where σ_i^e ($i = 1, 2, 3$) is the effective principal stress and A and B are empirical parameters. The Priest criterion is similar to the Pan–Hudson criterion in that it is circular on the π plane.

Later, Priest (2012) developed a simplified Priest criterion that can be expressed as:

$$\sigma'_{1hb} = \sigma'_{3hb} + \sigma_c \left(m \frac{\sigma'_{3hb}}{\sigma_c} + 1 \right)^{0.5}, \quad (10)$$

where

$$\begin{cases} \sigma'_{3hb} = w\sigma'_2 + (1-w)\sigma'_3 \\ \sigma'_{1hb} = \sigma'_1 + \sigma'_2 + \sigma'_3 - 2\sigma'_{3hb} = \sigma'_1 + (1-2w)\sigma'_2 + (2w-1)\sigma'_3, \\ w \approx \alpha\sigma'_3 \end{cases}$$

where σ'_i is the effective stress.

However, it should be noted that the Priest criterion cannot be reduced to the Hoek–Brown criterion, and the problems of smoothness and convexity still exist.

2.4 Benz Strength Criterion

Benz et al. (2008) proposed a revised Hoek–Brown criterion based on the spatial mobilized plane, which could properly consider the 3D property of rock material and expressed as Eq. (11).

$$f_{HB,MN} = q - LM_{c,HB}(p - p^*), \quad (11)$$

where

$$p^* = \begin{cases} p_t^s = -\frac{\sigma_{ci}^s}{m_b}, & \text{for } M_{c,HB} = M_{c,HB}^s \\ p_t^s = -\frac{\tilde{f}}{M_{c,HB}} - \frac{\tilde{f}}{3} + \sigma_3, & \text{for } M_{c,HB} = M_{c,HB}^t \end{cases}.$$

The calculation based on the Benz strength criterion cannot avoid the use of Lode dependences, which is in accordance with the modified GZZ strength criterion (Zhang et al. 2013).

2.5 Jiang Strength Criterion

Jiang (2017a, b), Jiang et al. (2011), Jiang and Xie (2012) and Jiang and Zhao (2015) developed several

three-dimensional Hoek–Brown criteria with various $g(\theta_\sigma)$ as shown in Table 1. These criteria have a common form that can be summarized as follows:

$$\frac{1}{(\sigma_c)^{(1/a-1)}} \left(\sqrt{3J_2} \right)^{1/a} + \frac{g(\theta_\sigma)}{\sqrt{3}} m \sqrt{J_2} - m \frac{I_1}{3} - s\sigma_c = 0. \quad (12)$$

The equation can be expressed on the π plane as follows:

$$\frac{1}{(\sigma_c)^{(1/a-1)}} \left(\frac{3}{\sqrt{2}} \tau_{oct} \right)^{1/a} + \frac{g(\theta_\sigma)}{\sqrt{2}} m \tau_{oct} - m \frac{I_1}{3} - s\sigma_c = 0. \quad (13)$$

2.6 GZZ Criterion

Mogi (2006) discovered a good relationship between τ_{oct} and $\sigma_{m,2}$ in the fit of triaxial compression strength data, and proposed the Mogi–Coulomb strength criterion (Mogi 1971b). Similarly, Zhang and Zhu (2007) proposed a three-dimensional Hoek–Brown criterion by combining the Hoek–Brown criterion with the Mogi criterion to yield the Zhang–Zhu strength criterion as follows:

$$\frac{9}{2\sigma_c} \tau_{oct}^2 + \frac{3}{2\sqrt{2}} m \tau_{oct} - m \sigma_{m,2} - s\sigma_c = 0. \quad (14)$$

Zhang modified the Zhang–Zhu criterion and developed a generalized Zhang–Zhu criterion, the GZZ criterion (Zhang 2008):

$$\frac{1}{(\sigma_c)^{(1/a-1)}} \left(\frac{3}{\sqrt{2}} \tau_{oct} \right)^{1/a} + \frac{m}{2} \left(\frac{3}{\sqrt{2}} \tau_{oct} \right) - m \sigma_{m,2} - s\sigma_c = 0, \quad (15)$$

where $\sigma_{m,2} = (\sigma_1 + \sigma_3)/2$. Substituting Eqs. (2) into (15), the GZZ criterion can be expressed in terms of:

Table 1 $g(\theta_\sigma)$ of three-dimensional Hoek–Brown criterion by Jiang

References	$g(\theta_\sigma)$	Main characteristics
Jiang et al. (2011), Jiang and Xie (2012)	$a_i + b_i \cos i \theta_\sigma$ ($i=1, 2, 3$)	there is a hypothetical form with no theoretical background related; criterion is smooth when $i=3$, and others are not
Jiang and Zhao (2015)	$2\cos\left(\frac{\pi}{3} - \theta_\sigma\right)$	Non-smooth
Jiang (2017a)	1.618–1.691	an average value of 1.65 is recommend by authors, which cannot be degenerates to Hoek–Brown criterion
Jiang (2017b)	$\frac{\cos\left[\frac{1}{3}\arccos(k\cos 3\theta_\sigma)\right]}{\cos\left[\frac{1}{3}\arccos(k)\right]}$	smooth and full convex criterion with additional parameter k ; complicated form and cannot be degenerates to Hoek–Brown criterion

$$\frac{1}{(\sigma_c)^{(1/a-1)}} \left(\frac{3}{\sqrt{2}} \tau_{\text{oct}} \right)^{1/a} + \left(\frac{3+2\sin\theta_\sigma}{2\sqrt{2}} \right) m \tau_{\text{oct}} - m \frac{I_1}{3} - s \sigma_c = 0. \quad (16)$$

To address the non-smoothness and non-convexity problems, Chen et al. (2021a) modified the GZZ criterion as follows:

$$\frac{1}{\sigma_c^{(1/a-1)}} \left(\frac{3}{\sqrt{2}} \tau_{\text{oct}} \right)^{1/a} + \frac{m_b}{2} \left(\frac{3}{\sqrt{2}} \tau_{\text{oct}} \right) - \frac{m_b}{2} \sqrt{\frac{18I_1^3 \tau_{\text{oct}}^2 - 81I_1^* \tau_{\text{ct}}^4 - 54I_3^* \tau_{\text{oct}}^2}{4I_1^3 - 18I_1^* \tau_{\text{oct}}^2 - 108I_3^*}} = 0, \quad (17)$$

where

$$\begin{cases} I_1^* = 3p^* = \sigma_1^* + \sigma_2^* + \sigma_3^* = I_1 + 3 \frac{\sigma_c}{m_b} s \\ I_3^* = \sigma_1^* \sigma_2^* \sigma_3^* = \left(\sigma'_1 + \frac{\sigma_c}{m_b} s \right) \left(\sigma'_2 + \frac{\sigma_c}{m_b} s \right) \left(\sigma'_3 + \frac{\sigma_c}{m_b} s \right) \end{cases}$$

The criterion developed by Chen et al. (2021a) has a complex mathematical expression compared to the original GZZ criterion.

The GZZ criterion degenerates into the Zhang–Zhu criterion when $a=0.5$. Further analysis found that the Pan–Hudson criterion (Eq. 8) is a special case at $\theta_\sigma=0$, and that the GZZ criterion degenerates to the Hoek–Brown criterion (Eq. 6) under triaxial compression and extension conditions, i.e. $\theta_\sigma = \pm \pi/6$. Additionally, the GZZ criterion possesses a high fitting accuracy to true triaxial rock strength data with reliable criterion parameters from Hoek's recommendation (Zhang 2008; Zhang and Zhu 2007).

3 A New GZZ Version

3.1 New Criterion in the Form of Stress Invariants

Further analysis found that the first two terms of the GZZ criterion (Eq. 15) are stress invariants. The distribution of σ_i ($i=1, 2, 3$) in stress invariants is symmetric, hence the non-smoothness of the GZZ criterion on the π plane is predominantly affected by the asymmetric term $\sigma_{m,2}$. Subsequently, we developed a new version of the GZZ criterion by modifying $\sigma_{m,2}$ to satisfy the smooth requirement on the π plane, and maintain the advantages associated with the GZZ criterion in terms of its degeneration to the Hoek–Brown criterion in triaxial compression and extension conditions. The modified $\sigma_{m,2}$ was relabeled as $(\sigma_{m,2})_{\text{new}}$, which should satisfy the boundary condition as follows:

$$(\sigma_{m,2})_{\text{new}} \Big|_{\theta_\sigma = \pm \frac{\pi}{6}} = (\sigma_{m,2})_{\text{GZZ}} \Big|_{\theta_\sigma = \pm \frac{\pi}{6}} = \frac{\sigma_1 + \sigma_3}{2}. \quad (18)$$

The corresponding new criterion can be expressed as:

$$\frac{1}{(\sigma_c)^{(1/a-1)}} \left(\frac{3}{\sqrt{2}} \tau_{\text{oct}} \right)^{1/a} + \frac{m}{2} \left(\frac{3}{\sqrt{2}} \tau_{\text{oct}} \right) - m(\sigma_{m,2})_{\text{new}} - s \sigma_c = 0. \quad (19)$$

In triaxial extension condition, i.e. $\theta_\sigma = -\pi/6$, the stress invariants are reduced to:

$$\begin{aligned} I_1 &= 2\sigma_1 + \sigma_3 \\ I_2 &= \sigma_1 \sigma_1 + 2\sigma_1 \sigma_3 \\ I_3 &= \sigma_1 \sigma_1 \sigma_3 \\ J_2 &= (\sigma_1 - \sigma_3)^2 / 3. \end{aligned} \quad (20)$$

In triaxial compression conditions, i.e. $\theta_\sigma = \pi/6$, the stress invariants are reduced to:

$$\begin{aligned} I_1 &= 2\sigma_3 + \sigma_1 \\ I_2 &= \sigma_3 \sigma_3 + 2\sigma_1 \sigma_3 \\ I_3 &= \sigma_3 \sigma_3 \sigma_1 \\ J_2 &= (\sigma_1 - \sigma_3)^2 / 3. \end{aligned} \quad (21)$$

It is a simple matter to prove that both cases, Eqs. (20) and (21), can be expressed in the form:

$$\sigma_1 \sigma_3 = \frac{6I_3 J_2}{I_1 I_2 - 9I_3}. \quad (22)$$

Substituting Eqs. (22) into (18), and noting that $I_3 = \sigma_1 \sigma_2 \sigma_3$, yields:

$$\begin{aligned} (\sigma_{m,2})_{\text{new}} \Big|_{\theta_\sigma = \pm \frac{\pi}{6}} &= \frac{\sigma_1 + \sigma_3}{2} = \frac{I_1 - \sigma_2}{2} = \frac{I_1 - I_3 / (\sigma_1 \sigma_3)}{2} \\ &= \frac{1}{2} \left(I_1 - \frac{I_1 I_2 - 9I_3}{6J_2} \right). \end{aligned} \quad (23)$$

Letting $(\sigma_{m,2})_{\text{new}} = \frac{1}{2} \left(I_1 - \frac{I_1 I_2 - 9I_3}{6J_2} \right)$, and noting that $J_2 = 3\tau_{\text{oct}}^2/2$ and $J_2 = (I_1^2 - 3I_2)/3$, the new criterion can be expressed as follows:

$$\begin{aligned} \frac{1}{(\sigma_c)^{(1/a-1)}} \left(\frac{3}{\sqrt{2}} \tau_{\text{oct}} \right)^{1/a} + \frac{m}{2} \left(\frac{3}{\sqrt{2}} \tau_{\text{oct}} \right) \\ - \frac{m}{2} \left(\frac{7}{6} I_1 + \frac{27I_3 - I_1^3}{27\tau_{\text{oct}}^2} \right) - s \sigma_c = 0. \end{aligned} \quad (24)$$

It should be noted that Eq. (24) may become singular when $\tau_{\text{oct}}=0$ under hydrostatic pressure conditions, i.e.

$\sigma_1 = \sigma_2 = \sigma_3$, which will be further discussed in the following section.

3.2 New Criterion on the π Plane

The relationships among θ_σ , stress invariants I_i and deviatoric invariants J_i can be expressed as follows:

$$\begin{cases} \sin 3\theta_\sigma = -\frac{3\sqrt{3}}{2} \cdot \frac{J_3}{J_2^{3/2}} \\ I_3 = -\frac{2}{3\sqrt{3}} J_2^{3/2} \sin 3\theta_\sigma - \frac{1}{3} I_1 J_2 + \frac{1}{27} I_1^3 \end{cases}, \quad (25)$$

where J_3 is the third deviatoric invariant, $J_3 = S_{ij}S_{jk}S_{ki}/3$, $S_{ij} = \sigma_{ij} - \sigma_{kk}\delta_{ij}/3$, and S_{ij} is the deviatoric stress tensor.

Substituting Eqs. (25) into (24), and noting that $J_2 = 3\tau_{\text{oct}}^2/2$, we can derive the new criterion on the π plane as follows:

$$\frac{1}{(\sigma_c)^{(1/a-1)}} \left(\frac{3}{\sqrt{2}} \tau_{\text{oct}} \right)^{1/a} + \left(\frac{3 + \sin 3\theta_\sigma}{2\sqrt{2}} \right) m \tau_{\text{oct}} - m \frac{I_1}{3} - s \sigma_c = 0. \quad (26)$$

As a further derivation of the new criterion (Eq. 24), Eq. (26) is suitable for any stress conditions with no calculation singularity.

Substituting $J_2 = 3\tau_{\text{oct}}^2/2$ into Eq. (26), we obtain another expression in the form of I_1, J_2, θ_σ as Eq. (27).

$$\frac{1}{\sigma_c^{(1/a-1)}} \left(\sqrt{3J_2} \right)^{1/a} + \left(\frac{3 + \sin 3\theta_\sigma}{2\sqrt{3}} \right) m_b \sqrt{J_2} - m_b \frac{I_1}{3} - s \sigma_c = 0. \quad (27)$$

Compared with the GZZ criterion (Eq. 16), the new criterion can also be written as:

$$\begin{aligned} \frac{1}{(\sigma_c)^{(1/a-1)}} \left(\frac{3}{\sqrt{2}} \tau_{\text{oct}} \right)^{1/a} + \left(\frac{3 + 2 \sin \theta_\sigma}{2\sqrt{2}} \right) m \tau_{\text{oct}} \\ - m \frac{I_1}{3} - s \sigma_c + \Delta \sigma_{m,2} m = 0. \end{aligned} \quad (28)$$

where $\Delta \sigma_{m,2} = (\sigma_{m,2})_{\text{new}} - (\sigma_{m,2})_{\text{GZZ}} = (\sin 3\theta_\sigma - 2 \sin \theta_\sigma) \tau_{\text{oct}} / 2\sqrt{2}$. The roots of equation $\Delta \sigma_{m,2} = 0$ are $\theta_\sigma = \pm \pi/6$ and $\theta_\sigma = 0$, hence the intersections of the new criterion with the GZZ criterion are $\theta_\sigma = \pm \pi/6$ and $\theta_\sigma = 0$, while its intersections with the Hoek–Brown criterion at $\theta_\sigma = \pm \pi/6$ are shown in Fig. 3. This indicates that the new criterion degenerates into the Hoek–Brown criterion under triaxial compression and extension conditions, which is in accordance with the GZZ criterion. Compared with the Hoek–Brown criterion and the Pan–Hudson criterion, the new criterion is much closer to the GZZ criterion in its

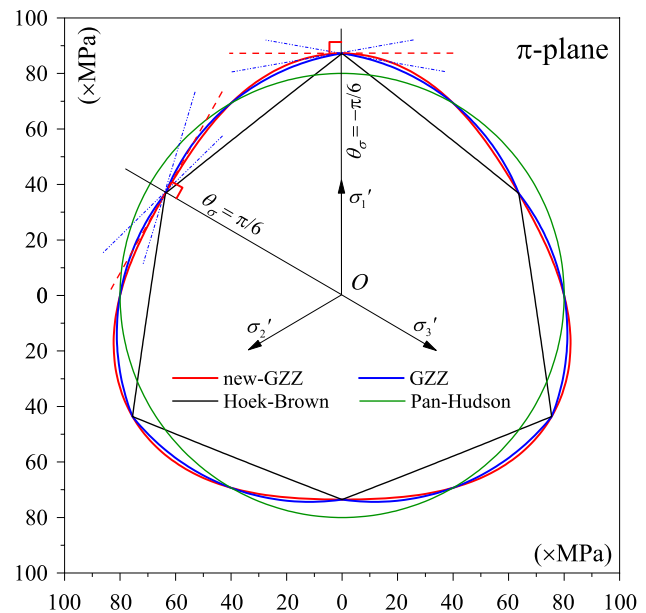


Fig. 3 π plane plot of the new GZZ criterion, GZZ criterion, Hoek–Brown criterion, and Pan–Hudson criterion at $I_1/3 = 120$ MPa for an average quality rock mass with $\sigma_c = 30$ MPa, $m = 4.5$, $s = 0.02$, $a = 0.5$

geometry on the π plane, making it reasonable to accept the new criterion as a new version of the GZZ criterion.

4 Smoothness and Convexity Examination of the New Strength Criterion on the π Plane

The criterion diameter vector to the origin point on the π plane is denoted by $r(\theta_\sigma)$, $r = \sqrt{2J_2} = \sqrt{3}\tau_{\text{oct}}$. The expression for the new criterion at a certain hydrostatic stress axis ξ on the π plane can be written as:

$$\begin{aligned} F\left(\xi = \frac{I_1}{\sqrt{3}}, r, \theta_\sigma\right) = \frac{1}{(\sigma_c)^{(1/a-1)}} \left(\frac{\sqrt{6}}{2} r \right)^{1/a} \\ + \left(\frac{3 + \sin 3\theta_\sigma}{2\sqrt{6}} \right) m r - m \frac{I_1}{3} - s \sigma_c = 0, \end{aligned} \quad (29)$$

where $\xi = I_1/3$ is the hydrostatic stress axis.

To avoid numerical singularity and iterative non-convergence in the rock strength calculation, the strength criterion should be smooth and convex on the π plane, which requires:

$$r'(\theta_\sigma = \pm \pi/6) = \left. \frac{dr}{d\theta_\sigma} \right|_{\theta_\sigma = \pm \pi/6} = 0. \quad (30)$$

As a deduction of Drucker's postulate (Drucker 1951), full convexity of a strength criterion on the π plane should satisfy:

$$r^2 + 2r'^2 - rr'' \geq 0. \quad (31)$$

4.1 Smoothness of the New Criterion

The total differential of $F(r, \theta)$ can be written as:

$$\frac{dF}{d\theta_\sigma} = \frac{\partial F}{\partial r} \frac{dr}{d\theta_\sigma} + \frac{\partial F}{\partial \theta_\sigma} = 0. \quad (32)$$

By substituting Eq. (29) into Eq. (32), we can obtain:

$$\begin{aligned} & \frac{(3/2)^{1/(2a)}}{a(\sigma_c)^{(1/a-1)}} r^{(1/a-1)} \times r'(\theta_\sigma) + \left(\frac{3 + \sin 3\theta_\sigma}{2\sqrt{6}} \right) m \times r'(\theta_\sigma) \\ & + \frac{\sqrt{6}}{4} m \times r \cos 3\theta_\sigma = 0. \end{aligned} \quad (33)$$

Simplification of Eq. (33) yields:

$$r'(\theta_\sigma) = \frac{-3r \cos 3\theta_\sigma}{3Ar^{(1/a-1)} + 3 + \sin 3\theta_\sigma}, \quad (34)$$

where $A = \frac{(3/2)^{1/(2a)}}{\sqrt{6}am(\sigma_c)^{(1/a-1)}/4}$ is a constant greater than 0. Although Eq. (34) is an implicit expression, $r'(\theta_\sigma = \pm\pi/6)|_{\text{new}} = 0$. The new criterion (Eq. 29) is thus smooth on the π plane.

$r'(\theta_\sigma) \leq 0$ is true for any stress conditions because $|\theta_\sigma| \leq \pi/6$ and $A > 0$. $r(\theta_\sigma)$ is thus a monotonic decreasing function with a continuous and derivable surface on the π plane for every θ_σ from $-\pi/6$ to $\pi/6$. It is simple to confirm that the stress diameter vector in the triaxial compression condition is larger than in the triaxial extension condition, i.e. $r(-\pi/6) > r(\pi/6)$. This strength property has been confirmed using triaxial test data for most rock types.

As for the GZZ criterion, $r'(\theta_\sigma = \pm\pi/6)|_{\text{GZZ}} = \frac{-2r \cos(\pm\pi/6)}{3Ar^{(1/a-1)} + 3 + 2 \sin(\pm\pi/6)} \neq 0$. The GZZ criterion can thus not satisfy the requirement of smoothness.

4.2 Convexity of the New Criterion

According to Eq. (34), the second derivative of $r(\theta_\sigma)$ can be obtained as follows:

$$r''(\theta_\sigma) = \frac{(1/a - 2)Ar^{(1/a-1)}r' \cos 3\theta_\sigma + 3[Ar^{(1/a-1)} + 1]r \sin 3\theta_\sigma - (\cos 3\theta_\sigma + \sin 6\theta_\sigma/6)r' + r}{[Ar^{(1/a-1)} + 1 + \sin 3\theta_\sigma/3]^2}. \quad (35)$$

By substituting the derived $r'(\theta_\sigma)$ and $r''(\theta_\sigma)$ into Eq. (31), we can generate the constraints of the new criterion that satisfy full convexity on the π plane as follows:

$$\tilde{r}^3 + k_1(\theta_\sigma) \times \tilde{r}^2 + k_2(\theta_\sigma) \times \tilde{r} + k_3(\theta_\sigma) \geq 0. \quad (36)$$

Here,

$$\tilde{r} = Ar^{(1/a-1)} \geq 0,$$

$$k_1(\theta_\sigma) = 2 - 7 \sin 3\theta_\sigma/3,$$

$$k_2(\theta_\sigma) = 2(1 + \sin 3\theta_\sigma/3)^2 - 6(1 + \sin 3\theta_\sigma/3) \sin 3\theta_\sigma + (1/a - 1) \cos^2(3\theta_\sigma),$$

$$k_3(\theta_\sigma) = (1 + \sin 3\theta_\sigma/3)^3 - 3(1 + \sin 3\theta_\sigma/3)^2 \sin 3\theta_\sigma.$$

Because θ_σ is the only variable of $k_i(\theta_\sigma)$, we can obtain the explicit expression $r = r(k_1, k_2, k_3) = r(\theta_\sigma)$. For all θ_σ from $-\pi/6$ to $\pi/6$, the inequality Eq. (36) is required for the criterion to be fully convex on the π plane. According to the geometric characteristics of the strength criterion on the π plane, the non-convexity appears predominantly in the triaxial extension condition. Therefore, to simplify the calculation process, the inequality of Eq. (36) must hold at $\theta_\sigma = \pi/6$ and is thus reduced to:

$$\tilde{r}^3 - \frac{1}{3}\tilde{r}^2 - \frac{40}{9}\tilde{r} - \frac{80}{27} \geq 0. \quad (37)$$

And, in triaxial compression conditions ($\theta_\sigma = -\pi/6$) the constraint condition should satisfy:

$$\tilde{r}^3 + \frac{13}{3}\tilde{r}^2 + \frac{44}{9}\tilde{r} + \frac{44}{27} \geq 0. \quad (38)$$

Inequality equation holds for all $r \geq 0$ ($\tilde{r} \geq 0$). The new criterion is, thus, convex for stress conditions near triaxial compression on the π plane, which is in accordance with other criteria such as HB strength criterion (Hoek et al. 2002), Pan–Hudson criterion (Pan and Hudson 1988), Priest criterion (Priest 2005) and Jiang criteria (Jiang 2017a, b; Jiang et al. 2011; Jiang and Xie 2012; Jiang and Zhao 2015), and results from many triaxial compression data (Yu 1992). However, it is known that the original Hoek–Brown criterion

was developed predominantly for those stress conditions near triaxial compression.

The cubic inequality Eq. (37) inequation has three solutions, but we exclude the two negative roots concerning the physical meaning of $\tilde{r} = Ar^{(1/a-1)} \geq 0$. By solving the inequality Eq. (37), we can obtain the following:

$$r \geq \left(\frac{5 + \sqrt{105}}{6A} \right)^{a/(1-a)} = \sigma_c \left[\frac{1.556am}{(3/2)^{1/(2a)}} \right]^{a/(1-a)}. \quad (39)$$

It is reasonable to treat $r_0 = \sigma_c \left[\frac{1.556am}{(3/2)^{1/(2a)}} \right]^{a/(1-a)}$ as a critical radius of the new criterion, causing the criterion to transition from non-convex to fully convex and becomes fully convex when $r \geq r_0$ at $\theta_\sigma = \pi/6$.

Equation (39) shows that the convexity of the new criterion is related to I_1 and r_0 on the π plane. I_1 reflects the stress level, and critical radius of the new criterion $r_0 = \sigma_c \left[\frac{1.556am}{(3/2)^{1/(2a)}} \right]^{a/(1-a)}$ reflects the physical and mechanical properties of rock, which is constant because the criterion parameters σ_c , m , and a are also constants for a certain engineering rock mass. The value of r corresponds to I_1 , which is governed by new-GZZ criterion as shown in Eq. (29). So, the convexity of the new criterion on the π plane for a certain rock mass is only related to I_1 . Additionally, an increase in I_1 will result in the transformation of the criterion from non-convex to fully convex, which is in accordance with the literature (Jiang et al. 2011). It is worth noting that the authors described the phenomenon qualitatively from a geometrical perspective, but this study provided a theoretical and mathematical explanation of this phenomenon.

4.3 Geometrical Analysis of the New Criterion for Rock Masses of Varying Quality

In the following geometrical analysis of the new criterion on the π plane, we predominantly concentrated on its smoothness and convexity for rock masses of different quality. The parameter settings for three different rock masses, i.e. a massive strong rock mass, an average quality rock mass, and a poor-quality rock mass, were all taken from Hoek and Brown's recommendations (Hoek and Brown 1997), which are shown in Table 2:

Table 2 Input parameters of different types of rock mass (Hoek and Brown 1997)

Type of rock mass	Input parameters of new-GZZ			
	σ_c (MPa)	m	a	s
Massive strong quality	110	7.25	0.5	0.062
Average quality	30	4.50	0.5	0.02
Poor quality	7.5	0.55	0.55	0

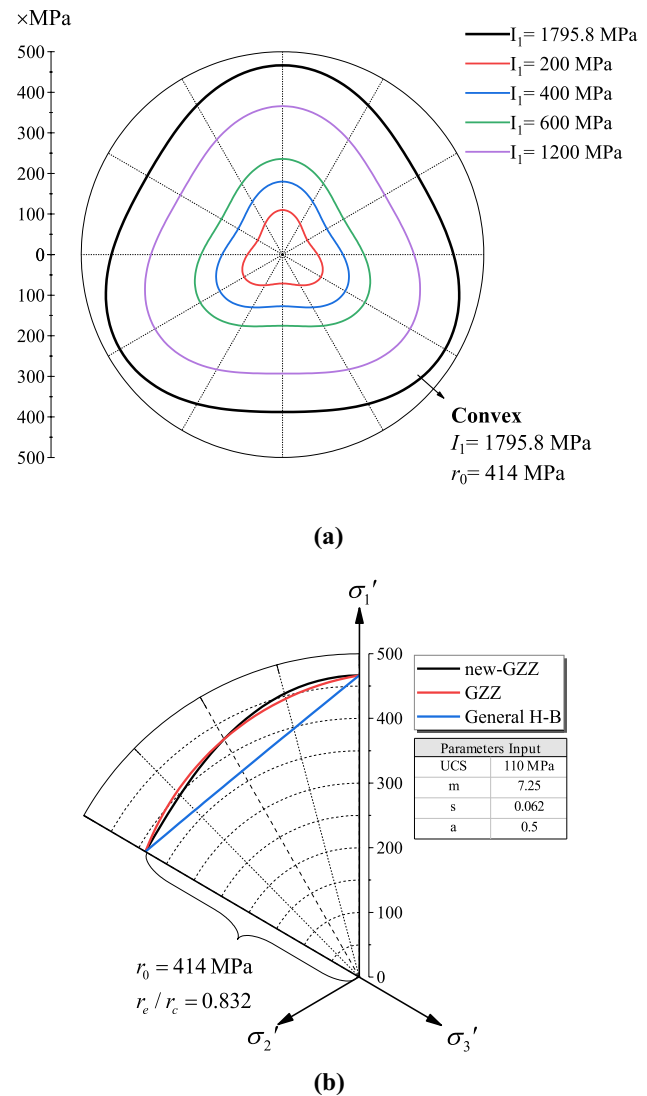


Fig. 4 **a** New criterion for massive strong rock mass at various I_1 ; and **b** new criterion, GZZ criterion, and general Hoek–Brown criterion for massive strong rock mass at critical radius (r_0 or I_1)

The calculation results are plotted in Figs. 4, 5, and 6, where Figs. 4a, 5a, and 6a are plots of the new criterion for different hydrostatic pressures $\xi = I_1/3$ on the π plane, and Figs. 4b, 5b, and 6b are plots of the new criterion, the GZZ criterion, and the general Hoek–Brown criterion, respectively, at the critical radius r_0 , which was calculated using Eq. (39).

The new criterion is smooth and derivable on the whole π plane for rock masses of various qualities, which is consistent with the previous theoretical results and avoids the numerical singularity in strength calculation. In addition, the new criterion perpendicularly intersects with the principal stress coordinate axis, which is also an important feature of criteria such as the Priest criterion (Priest and Engineering

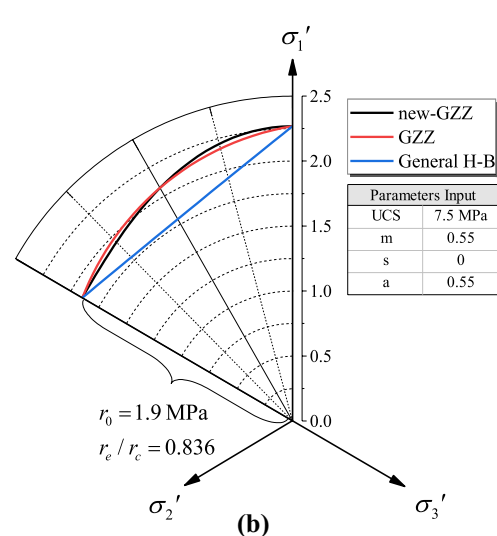
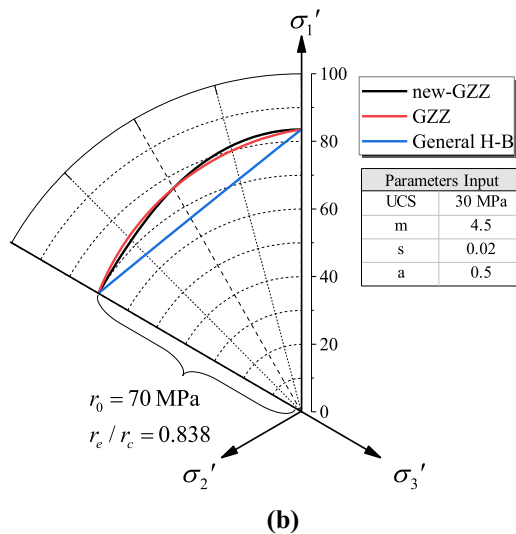
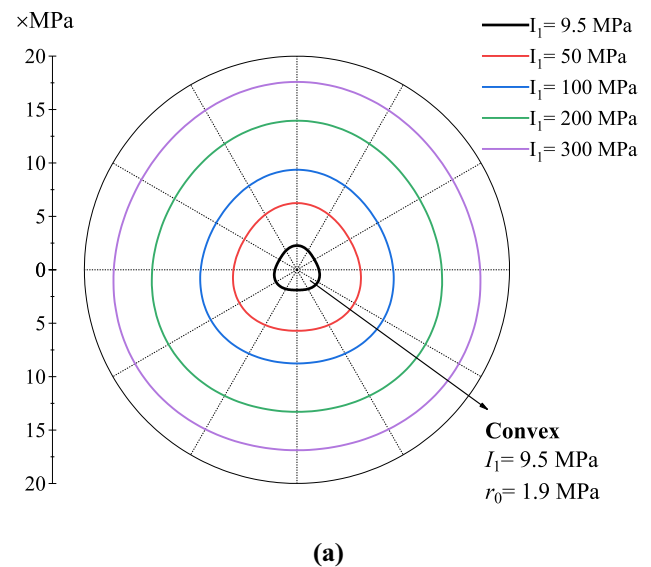
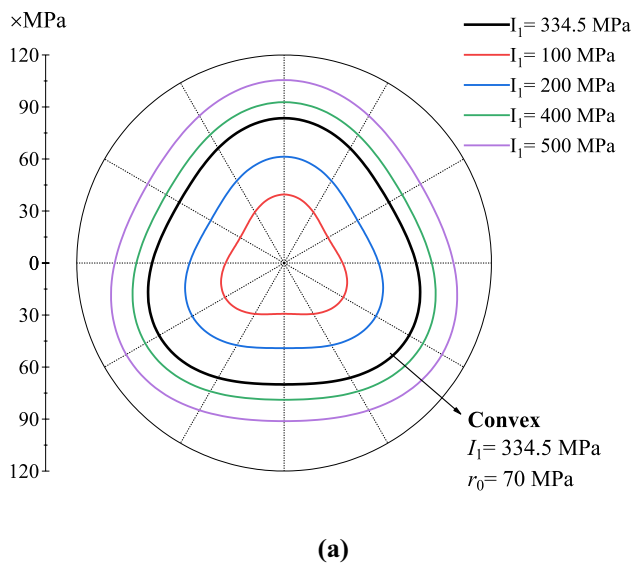


Fig. 5 **a** New criterion for average quality rock mass at various I_1 ; and **b** new criterion, GZZ criterion, and general Hoek–Brown criterion for average quality rock mass at critical radius (r_0 or I_1)

Fig. 6 **a** New GZZ criterion for poor quality rock mass at various I_1 ; and **b** new criterion, GZZ criterion, and general Hoek–Brown criterion for poor quality rock mass at critical radius (r_0 or I_1)

2005), Pan–Hudson criterion (Pan and Hudson 1988), and the unified strength criterion (You 2009).

For these rock masses of varying quality, the new criterion is not convex near the triaxial extension condition ($\theta_\sigma = \pi/6$) and when I_1 is small. It is commonly known that the Hoek–Brown criterion was originally developed for those rocks near the triaxial compression condition, a condition in which the new criterion is convex. With an increase in I_1 , the new criterion gradually changes from non-convex to fully convex at $r_0 = \sigma_c [1.556am/(3/2)^{1/(2a)}]^{a/(1-a)}$. The new criterion is thus fully convex when I_1 is relatively large. For rock masses of poor quality, the new criterion is fully convex even when I_1 is small. For various other rock masses, the ratio of the radius at $\theta_\sigma = \pi/6$ to $\theta_\sigma = -\pi/6$ at the critical r_0 , i.e. $\beta = r(\pi/6)/r(-\pi/6)$, are all in the range 0.83–0.84.

According to the new criterion (Eq. 23), and when a is equal to 0.5, it follows that:

$$r = \sqrt{3}\tau_{\text{oct}} = \frac{-\left(\frac{3+\sin 3\theta_\sigma}{2\sqrt{2}}\right)m\sigma_c + \sigma_c \sqrt{\left(\frac{3+\sin 3\theta_\sigma}{2\sqrt{2}}\right)^2 m^2 + \left(\frac{6mI_1}{\sigma_c} + 18s\right)}}{3\sqrt{3}} \quad (40)$$

So,

$$\beta = r\left(\frac{\pi}{6}\right)/r\left(-\frac{\pi}{6}\right) = \frac{-2 + \sqrt{4 + 12\left(\frac{I_1}{m\sigma_c} + \frac{3s}{m^2}\right)}}{-1 + \sqrt{1 + 12\left(\frac{I_1}{m\sigma_c} + \frac{3s}{m^2}\right)}} \quad (41)$$

According to the mathematical analysis, β increases with I_1 and β varies from 0.5 to 1. When I_1 is high enough, the new criterion is approximately circular on the π plane (β approaches 1), and in that case, the intermediate principal stress σ_2 has little influence on the rock strength because the value of β also reflects the effect of σ_2 on rock strength. Therefore, with an increase in I_1 , the effect of σ_2 on rock strength decreases, which has been proved by extensive triaxial compression tests. The critical I_1 , corresponding to r_0 , decreases as the quality of the engineering rock decreases. From Fig. 6b it can be seen that the new criterion is smooth and fully convex when I_1 is greater than 9.5 MPa for poor-quality rock masses with a σ_c of 7.5 MPa.

Based on this analysis, the new criterion is smooth on the whole π plane and satisfies the fully convex requirement for rock masses of poor quality or rock masses with slightly higher I_1 .

4.4 Plastic Flow Vector of the New Criterion

The plastic flow vector based on the new criterion is the core research of a strength criterion and also the basis of finite element calculation. Considering the associated flow rule, the plastic potential function and yield criterion have the same mathematical form as

$$Q = F = \frac{1}{\sigma_c^{(1/a-1)}} (\sqrt{3}J_2)^{1/a} + \left(\frac{3 + \sin 3\theta_\sigma}{2\sqrt{3}} \right) m_b \sqrt{J_2} - m_b \frac{I_1}{3} - s\sigma_c. \quad (42)$$

The plastic flow vector ($\partial Q / \partial \sigma$) can be written as:

$$\left[\frac{\partial Q}{\partial \sigma_1} \quad \frac{\partial Q}{\partial \sigma_2} \quad \frac{\partial Q}{\partial \sigma_3} \right] = \left[\frac{\partial Q}{\partial I_1} \quad \frac{\partial Q}{\partial \sqrt{J_2}} \quad \frac{\partial Q}{\partial J_3} \right] [T]_{3 \times 3}, \quad (43)$$

where

$$\begin{cases} \frac{\partial Q}{\partial I_1} = -\frac{1}{3}m_b \\ \frac{\partial Q}{\partial \sqrt{J_2}} = \frac{\sqrt{3}}{a\sigma_c^{(1/a-1)}} (\sqrt{3}J_2)^{1/a-1} + \left(\frac{\sqrt{3}}{2} - \frac{\sin 3\theta_\sigma}{\sqrt{3}} \right) m_b \\ \frac{\partial Q}{\partial J_3} = -\frac{3m_b}{4J_2} \end{cases} \quad (44)$$

and

$$[T]_{3 \times 3} = \begin{bmatrix} 1 & 1 & 1 \\ S_1/(2\sqrt{J_2}) & S_2/(2\sqrt{J_2}) & S_3/(2\sqrt{J_2}) \\ S_2S_3 + J_2/3 & S_1S_3 + J_2/3 & S_1S_2 + J_2/3 \end{bmatrix}.$$

The plastic flow vector $\partial Q / \partial \sigma$ can be computed using Eq. (43) which is simpler than the original GZZ, and $\sin 3\theta_\sigma$ in the term of $\partial Q / \partial \sqrt{J_2}$ is zero when $\theta_\sigma = \pm \pi/6$. This proves

that the plastic flow vector of new criterion is continuous and derivative.

5 Comparison of Fitting Precision of Pan–Hudson Criterion, GZZ Criterion, and the New Criterion

5.1 Materials and Fitting Index

Reliable and abundant triaxial compression test data is essential for evaluating the fitting accuracy of a criterion. In this study, we collected more than 30 sets of experimental data from various studies. However, some of the data had no associated σ_c data, which is a necessary parameter for those criteria developed based on the Hoek–Brown criterion. We thus used the remaining 15 data sets containing σ_c for our calculations, as shown in Table 3.

The triaxial strength data was inevitably discrete because of its heterogeneity and anisotropy, which may cause large errors when the least-squares method is applied. In our study, we used the sum of absolute error (SAE) method to fit the data, as suggested by You (2010a, b), which provides the mean least absolute error and can effectively be used to reduce the influence of abnormal data. We could use mf to evaluate the fitting accuracy of a criterion as follows:

$$mf = \min \left\{ \sum_{i=1}^N |(\tau_{\text{oct}})_i - f[(\sigma_{m,2})_i]| / N \right\}. \quad (45)$$

5.2 Fitting Analysis of Three Strength Criteria

Figure 7 shows the fitting of the new criterion, the GZZ criterion, and the Pan–Hudson criterion for 15 different types of intact rock. From this, we could make the preliminary observation that the difference between the new criterion and the GZZ criterion was tiny in the space of $\tau_{\text{oct}} - \sigma_{m,2}$, in particular for SAFOD granodiorite, Bentheim sandstone, Soignies limestone, Inada granite, and Mizuho trachyte.

The fitting results mf and R^2 are listed in Table 4.

Hoek and Brown (1997) and Marinos and Hoek (2001) recommended the value of the parameter m for the Hoek–Brown criterion according to many triaxial rock strength data as shown in Table 4. From the results we can deduce that only two values of m from the new criterion are outside of the recommended values for the 15 groups of different rock strength data. Among these, the Coconino sandstone m value is slightly higher than recommended, and the results from the new criterion and the GZZ criterion for SAFOD granodiorite are both far from the recommended range, and this may result from the significant variations in

Table 3 Data source for analysis

No.	Rock	σ_c (MPa)	Author(s)
1	SAFOD granodiorite	164	Lee and Haimson (2011), Haimson (2011)
2	Coconino sandstone	56.1	Ma (2014)
3	Bentheim sandstone	30	Ma (2014)
4	Yamaguchi marble	82	Mogi (1971a, b, 2006)
5	Dunham dolomite	265	Mogi (1971a, b, 2006)
6	Solnhofen limestone	310	Mogi (1971a, b, 2006)
7	Soignies limestone	139	Descamps and Tshibangu (2007)
8	Mizuho trachyte	100	Mogi (1971a, b, 2006)
9	KTB amphibolite	165	Chang and Haimson (2000)
10	Westerly granite	201	Haimson and Chang (2000)
11	Inada granite	226	Mogi (1971a, b, 2006)
12	Apache Leap tuff	147	Wang and Kemeny (1995)
13	Manazuru andesite	140	Mogi (1971a, b, 2006)
14	Orikake monzonite	234	Mogi (1971a, b, 2006)
15	Maha Sarakham salt	23	Sriapai et al. (2013)

the mechanical properties of SAFOD granodiorite. However, the GZZ criterion generated five values outside of the recommended range, and most of the remaining values were close to the upper limit of the recommended values, and this may result in a slightly conservative stability analysis of rock masses when directly using Hoek's recommendations. The parameters fitted using the Pan–Hudson criterion were far beyond the range. The parameters from the Hoek–Brown criterion can thus not be directly used for the Pan–Hudson criterion. Additionally, from the relative error (shown as RE in Table 4) between the values of m fitted by those three criteria and by 2D Hoek–Brown criterion showed that new-GZZ and GZZ exhibited much lower relative error compared with that of Pan–Hudson criterion, and this also verified the applicability and reliability of GZZ and new-GZZ for using the same parameters as the Hoek–Brown criterion for strength calculation.

The quantitative evaluation results of three different strength criteria using mf and R^2 are compared in Fig. 8. The fitting accuracy of the new criterion was typically similar to that of the GZZ criterion, with high R^2 and small mf , using the rock strength data from 15 different rock types, and this indicated the modification of smoothed new-GZZ based on the original GZZ will not cause the decrease of accuracy in fitting the rock strength data. The fitting accuracy was much higher than that of the Pan–Hudson criterion, which further proves that the new criterion and the GZZ criterion are applicable for strength calculations. It is worth noting that the fitted parameter m was based on the fit of the intact rock strength data with high m_i , which was much smaller than m_b for the engineering rock mass. Based on the foregoing analysis, the new criterion is more adaptable for rock masses of poor quality, which indicates that it may be more reliable when used for engineering rock masses. Besides, the

best method to evaluate the adaptability of a criterion is to study its description accuracy for the test data, and higher fitting accuracies of new-GZZ for 13 kinds of rocks under various stress conditions indicated the wider adaptability scopes of the criterion. This study further indicated that the new-GZZ is applicable especially for the deep-buried rock masses under high geostresses, showing the characteristics of soft rock in shear stress conditions.

6 Conclusion

This study involved the development of a new version of the GZZ criterion to solve the original GZZ criterion's non-smoothness and non-convexity on the π plane. The geometric characteristics on the π plane, applicability for rock masses, and fitting accuracy for rocks were discussed and the following conclusions were drawn:

- (1) By replacing $\sigma_{m,2}$ in the GZZ criterion with an alternative term $(\sigma_{m,2})_{\text{new}} = \frac{1}{2} \left(I_1 - \frac{I_1 I_2 - 9 I_3}{6 J_2} \right)$, we developed a new version of the GZZ criterion in terms of stress invariants with smooth and derivable geometry. Unlike the original GZZ criterion, the new criterion did not have a complex form when translated into the form $F(I_1, J_2, \theta_\sigma) = 0$ on the π plane, and the plastic flow vector of new criterion $(\partial Q / \partial \sigma)$ was proved to be continuous and derivative. This avoided a numerical singularity in the strength calculation and also the complicated calculation process caused by the modified GZZ criterion incorporating an additional Lode dependence function.

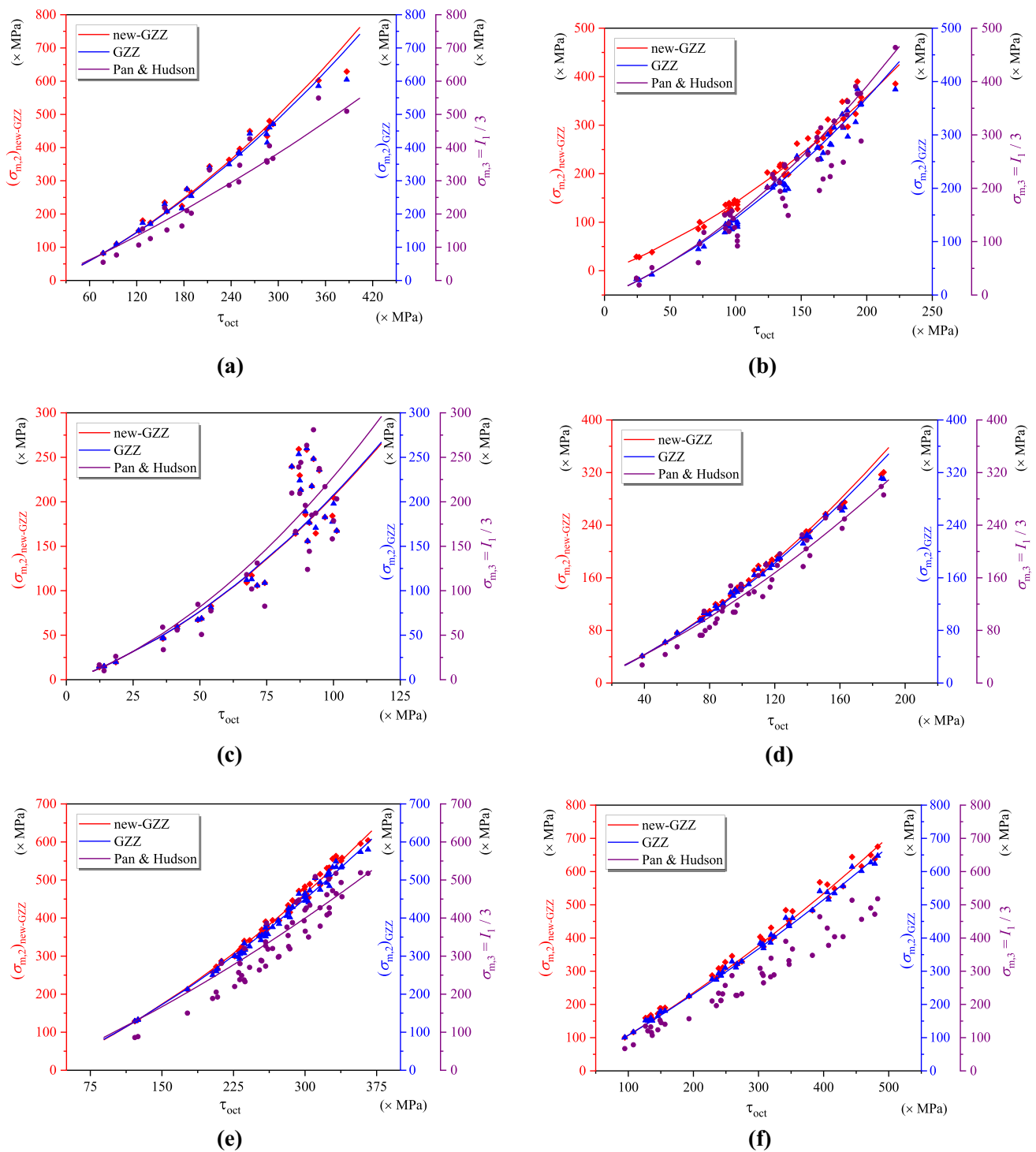


Fig. 7 Fitting results of the three strength criteria for various rock types: **a** SAFOD granodiorite, **b** Coconino sandstone, **c** Bentheim sandstone, **d** Yamaguchi marble, **e** Dunham dolomite, **f** Solnhofen

limestone, **g** Soignies limestone, **h** Mizuho trachyte, **i** KTB amphibolite, **j** Westerly granite, **k** Inada granite, **l** Apache Leap tuff, **m** Manazuru andesite, **n** Orikake monzonite, and **o** Maha Sarakham rock salt

- (2) As I_1 increased, the new criterion gradually changed from non-convex to fully convex. The critical transformation radius r_0 was linearly related to σ_c as follows: $r_0 = \sigma_c [1.556am/(3/2)^{1/(2a)}]^{a/(1-a)}$, which is a constant

for a particular engineering rock mass, just as σ_c , m , and a are constants. When analyzing the geometric characteristics of the new criterion for three different types of rock masses: good quality, average quality, and

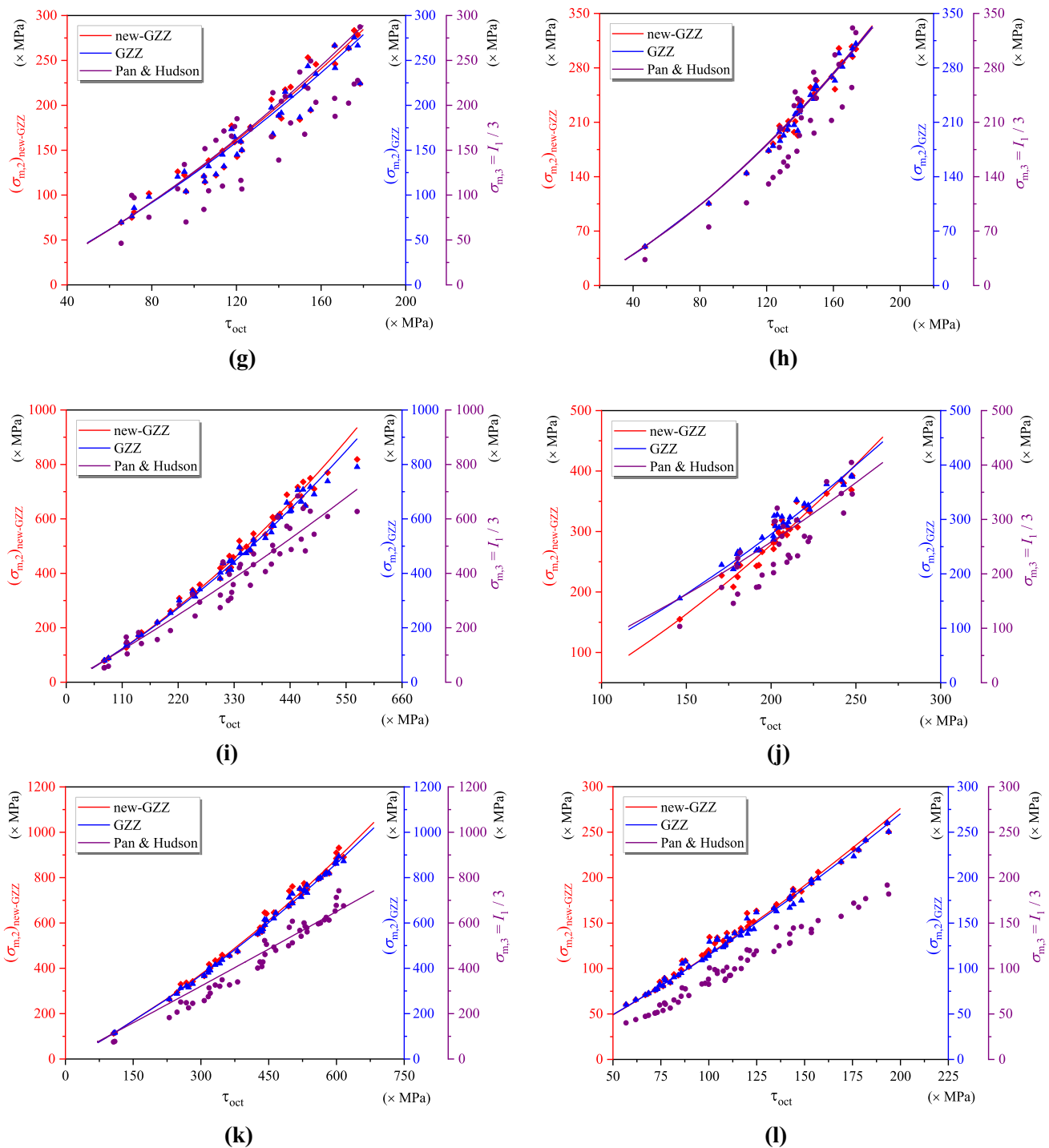


Fig. 7 (continued)

poor quality, we found that the new criterion was convex for those stress conditions with relatively large I_1 and for poor quality rock masses. Moreover, the Hoek–Brown criterion was originally developed for those rock masses near the triaxial compression condition, and our new criterion was also suitable for this condition.

- (3) We analyzed the fitting precision and reliability of the criterion parameters of the new criterion, GZZ criterion, and Pan–Hudson criterion using the minimum deviation fitting method with mf as the fitting index. We found that both the new criterion and the GZZ criterion had high fitting accuracy for 15 different rock types,

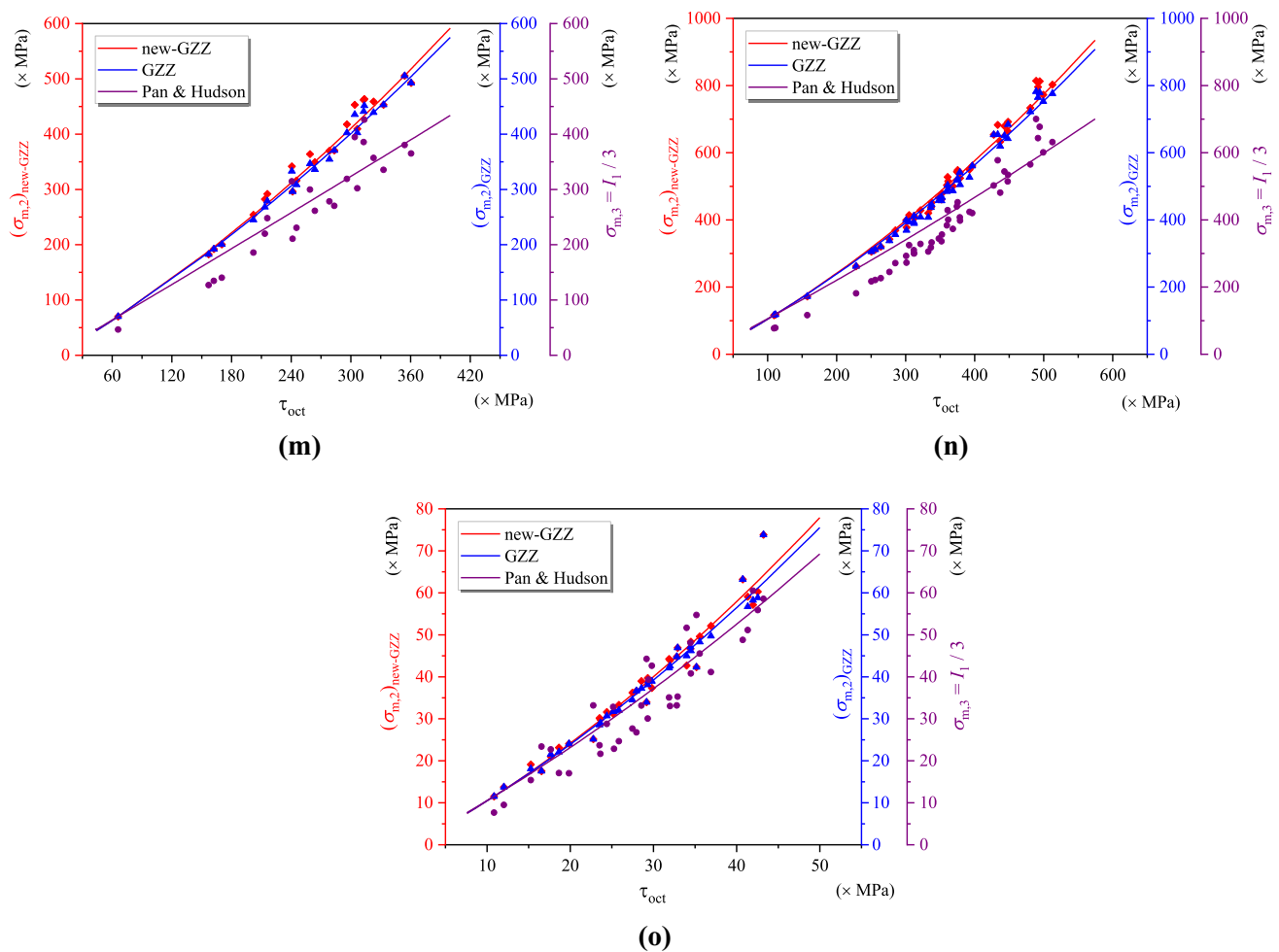


Fig. 7 (continued)

and their fitting accuracy was much higher than that of the Pan–Hudson criterion. Further analysis found that the parameters of the new criterion were closer to Hoek’s recommendation when compared with those of the GZZ criterion and the Pan–Hudson criterion could not directly adopt the recommended parameters for

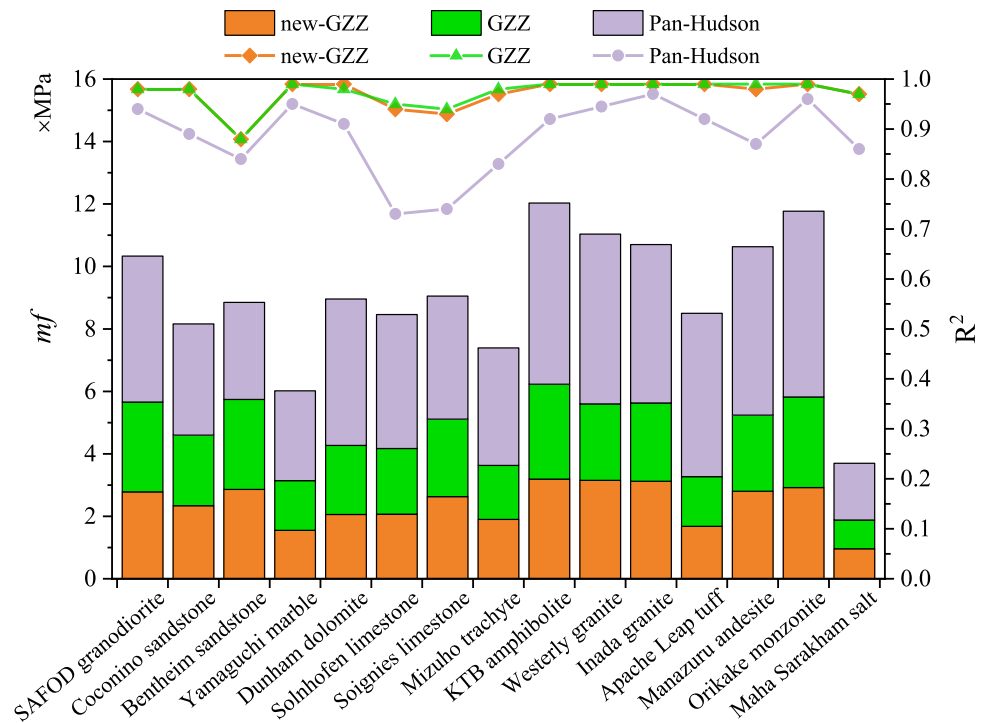
strength calculation because of its complete irrelevance with respect to recommended values. This proved that the new criterion was more applicable and reliable in terms of the calculation of rock strength when using the same parameters as Hoek–Brown criterion.

Table 4 Fitting precisions of new-GZZ, GZZ and Pan–Hudson strength criteria for different rocks

Rock	σ_c (MPa)	m^a	$m^{[HB]}$	new-GZZ			GZZ			Pan-Hudson					
				m^b	RE ^g (%)	mf (MPa)	R^2	m^b	RE (%)	mf (MPa)	R^2	m^b	RE (%)	mf (MPa)	R^2
SAFOD granodiorite	164	29 ± 3	12.91 ^f	12.9	0.08	2.78	0.98	13.8	6.89	2.88	0.98	35.7	176.53	4.67	0.94
Coconino sandstone	56.1	17 ± 4	23.21	21.5	7.37	2.34	0.98	20.1	13.40	2.26	0.98	17.6	24.17	3.56	0.89
Bentheim sandstone	30	17 ± 4	15.15	14.7	2.97	2.86	0.88	14.5	4.29	2.88	0.88	12.0	20.79	3.11	0.84
Yamaguchi marble	82	9 ± 3	10.51	11.9	13.23	1.55	0.99	12.9	22.74	1.59	0.99	17.7	68.41	2.88	0.95
Dunham dolomite	265	9 ± 3	9.59	8.7	9.28	2.06	0.99	9.6	0.10	2.21	0.98	15.5	61.63	4.69	0.91
Solnhofen limestone	310	None ^d	4.22	4.1	2.84	2.07	0.94	4.5	6.64	2.10	0.95	5.8	37.44	4.29	0.73
Soignies limestone	139	9 ± 2	10.37	9.7	6.46	2.63	0.93	10.4	0.29	2.48	0.94	9.3	10.32	3.94	0.74
Mizuho trachyte	100	13 ± 4	11.23	10.1	10.06	1.90	0.97	10.2	9.17	1.73	0.98	10.3	8.28	3.76	0.83
KTB amphibolite	165	26 ± 6	34.1	26.7	21.70	3.19	0.99	30.4	10.85	3.04	0.99	85.6	151.03	5.80	0.92
Westerly granite	201	32 ± 3	38.1	30.9	18.90	3.15	0.99	37.1	2.62	2.45	0.99	Singularity ^e			
Inada granite	226	32 ± 3	30.59	28.4	7.16	3.12	0.99	30.7	0.36	2.51	0.99	523.4	1611.02	5.07	0.97
Apache Leap tuff	147 ^c	13 ± 5	17.92	16.9	5.69	1.68	0.99	18.5	3.24	1.59	0.99	Singularity ^e			
Manazuru andesite	140	25 ± 5	34.71	29.9	13.86	2.80	0.98	33.2	4.35	2.44	0.99	537.0	1447.10	5.39	0.87
Oriskane monzonite	234 ^c	None ^d	19.36	18.8	2.89	2.92	0.99	20.5	5.89	2.90	0.99	67.2	247.11	5.95	0.96
Maha Sarakham salt	23	None ^d	15.05	18.7	24.25	0.96	0.97	20.7	37.54	0.92	0.97	28.8	91.36	1.82	0.86

^a According to Hoek (Hoek and Brown 1997; Marinatos and Hoek 2001), ^b best fitting, ^c average value, ^d not find, ^e totally unreasonable fitting results, ^f data with gray background means inconsistent with Hoek's recommendation, [HB] best fitting by 2D Hoek–Brown criterion, ^g relative error and $RE = |m^b - m^{[HB]}| / m^{[HB]}$

Fig. 8 Comparison of the fitting precision of three strength criteria for 15 different rock types



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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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